

INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
 in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

WORKED SOLUTIONS

CLASS

INDEX NUMBER

CHEMISTRY

9729/04

Paper 4 Practical

15 Aug 2017

Candidates answer on the Question Paper.

2 hours 30 minutes

READ THESE INSTRUCTIONS FIRST

Write your index number, name and class on all the work you hand in.
 Give details of the practical shift and laboratory when appropriate, in the boxes provided.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

Qualitative Analysis Notes are printed on pages 18 and 19.

At the end of the examination, fasten all your work securely together.
 The number of marks is given in the brackets [] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	17
2	24
3	14
Total	55

This document consists of **15** printed pages.



- 1 In this experiment you will determine the relative atomic mass, A_r , of magnesium by a titration method.

FA 1 is 2.00 mol dm^{-3} hydrochloric acid, HCl .

FA 3 is $0.120 \text{ mol dm}^{-3}$ sodium hydroxide, NaOH

magnesium ribbon

bromophenol blue indicator

(a) Method

Reaction of magnesium with **FA 1**

- Pipette 25.0 cm^3 of **FA 1** into the 250 cm^3 beaker.
- Weigh the strip of magnesium ribbon and record its mass.

mass of magnesium =g

- Coil the strip of magnesium ribbon loosely and then add it to the **FA 1** in the beaker.
- Stir the mixture occasionally and wait until the reaction has finished.

Dilution of the excess acid

- Transfer all the solution from the beaker into the volumetric flask.
- Make up the solution up to the mark using distilled water.
- Shake the flask to mix the solution before using it for your titrations.
- Label this solution of hydrochloric acid **FA 2**.

Titration

- Fill the burette with **FA 2**.
- Rinse the pipette out thoroughly. Then pipette 25.0 cm^3 of **FA 3** into a conical flask.
- Add several drops of bromophenol blue indicator.
- Run **FA 2** from the burette into this flask until the mixture just becomes yellow.
- Carry out as many titrations as you think necessary to obtain consistent results.
- Make certain that your recorded results show the precision of your working.
- Record in a suitable form on below all your burette readings and the volume of **FA 2** added in each accurate titration.

Results

The following readings are recorded

- Mass of Mg used
- Initial and final burette reading for two (or more) accurate titrations **M1**

Titre values recorded for accurate titrations and appropriate headings for the accurate titration tables and cm^3 units.

- Initial (burette) reading
- Final (burette) reading
- Titre or volume of **FA 2** used/added (*not "difference" or "total"*)
- Unit: cm^3 (for each heading)

*If cm^3 units are not given in the headings, every entry in the table must have the correct unit. **M2***

All accurate burette reading are to the nearest 0.05 cm^3 . Do **not** award this mark if:

- 50.00cm³ is used as an initial burette reading
- Any burette reading is greater than 50.00cm³
- There is only one accurate titration **M3**

There are two uncorrected **accurate** titres within 0.10 cm³. *Do not award this mark if, having performed two titre within 0.10 cm³ a further titration is performed.* **M4**

Calculate the student's mean titre based on the following hierarchy:

- Value of 2 identical titres
- Average of titres within 0.05 cm³
- Average of titres within 0.1 cm³

Award based on the difference, Δtitre , between Student's and Supervisor's mean titre. Give 2 marks if $\Delta\text{titre} \leq 0.20 \text{ cm}^3$

Give 1 marks if $0.20 < \Delta\text{titre} \leq 0.50 \text{ cm}^3$ **M5,M6**

[6]

- (b) From your titrations, obtain a suitable volume of **FA 2** to be used in your calculations. Show clearly how you obtained this volume.

Student obtains appropriate "average" to 2 d.p., from any experiments with uncorrected end – point titre values within 0.10 cm³

Do not award this mark if the titres used are not identified either in the table (by, for example, a tick) or in a calculation.

*Do **not** award this mark if there are arithmetic error in the table.* **M7**

Volume of **FA 2** =[1]

(c) Calculations

Show your working and appropriate significant figures in the final answer to **each** step of your calculations.

Units should be either in final calculation **or** dotted line.

Sf will be deducted ½ for overall, max 1 mark

Units will be deducted ½ for overall, max 1 mark

Working must be shown for all parts.

- (i) Deduce the amount of hydrochloric acid in the volume of **FA 2** you calculated in (b)

$$\text{Amount of NaOH} = 0.120 \times \frac{25.0}{1000} = 0.003000 \text{ mol}$$



$$\begin{aligned} \text{Amount of HCl} &= \text{amount of NaOH} \\ &= 0.00300 \text{ mol} \end{aligned} \quad \mathbf{M8}$$

Amount of HCl = [1]

- (ii) In (a), you reacted 25.0 cm³ of **FA 1** with your weighed piece of magnesium. After the reaction, the unreacted hydrochloric acid was used to prepare 250 cm³ of **FA 2**.

Calculate the amount of hydrochloric acid that reacted with the magnesium ribbon.

$$\text{Amount of HCl in volumetric flask} = 0.00300 \times \frac{250}{(b)} = \mathbf{a} \text{ mol}$$

$$\text{Amount of HCl used initially} = 2.00 \times \frac{25.0}{1000} = 0.0500 \text{ mol} \quad \mathbf{M9}$$

$$\begin{aligned} \text{Amount of hydrochloric acid that reacted with Mg} &= 0.0500 - \mathbf{a} \\ &= \mathbf{b} \text{ mol} \quad \mathbf{M10} \end{aligned}$$

$$\text{Amount of HCl reacting with Mg} = \dots\dots\dots[2]$$

(iii) Hence, calculate the relative atomic mass, A_r , of magnesium.



$$\text{Amount of Mg} = \mathbf{b} / 2 \text{ mol}$$

$$A_r = \text{mass of Mg} / (\mathbf{b}/2)$$

$$= \frac{2 \times \text{mass of Mg}}{\mathbf{b}} \quad (1\text{dp}) \quad \mathbf{M11}$$

Must be 1dp otherwise 0 mark.

$$A_r \text{ of Mg} = \dots\dots\dots[1]$$

(d) A student carried out the same experiment but used 1.00 g of magnesium ribbon. State and explain why the student's experiment could not be used to determine the value for the A_r of magnesium. Include a calculation in your answer.

[A_r : Mg, 24.3]

$$\text{Amount of Mg} = 1/24.3 = 0.0412 \text{ mol}$$

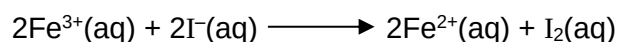
$$\text{Amount of HCl needed} = 0.0412 \times 2 = 0.0824 \text{ mol but only 0.05 mol present} \quad \mathbf{M12}$$

If 1.0g of Mg is used, Mg would be in excess/ acid would be the limiting reagent/ all the acid would be used up. $\mathbf{M13}$

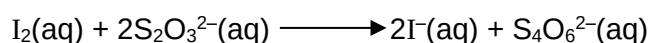
[2]

[Total: 13]

2 You will investigate the rate of reaction between iron(III) ions, Fe^{3+} , and iodide ions, I^- .



The iodine, I_2 , produced can be reacted immediately with thiosulfate ions, $\text{S}_2\text{O}_3^{2-}$.



When all the thiosulfate has been used, the iodine produced will turn starch indicator blue-black.

The rate of the reaction can therefore be measured by finding the time for the blue-black colour to appear.

FA 4 is aqueous iron(III) chloride, FeCl_3 .

FA 5 is aqueous potassium iodide, KI.

FA 6 is $0.0060 \text{ mol dm}^{-3}$ sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3$.
starch indicator

You are advised to read the instructions before starting any practical work and draw a table for your results in the space on page 6.

(a) Method

Experiment 1

- Fill a burette with **FA 4**.
- Run 20.00 cm^3 of **FA 4** into a 100 cm^3 beaker.
- Use the measuring cylinder to place the following in a second 100 cm^3 beaker.
 - 10 cm^3 of **FA 5**
 - 20 cm^3 of **FA 6**
 - 10 cm^3 of starch indicator
- Add the contents of the second beaker to the first beaker and start timing.
- Stir the mixture once and place the beaker on the white tile.
- The mixture turns brown and then yellow before turning a blue-black colour. Stop timing when this **blue-black colour** appears.
- Record in your table the volume of **FA 4** used, the volume of distilled water used and the time to the **nearest second** for the blue-black colour to appear.
- Wash both beakers.

For each of **Experiments 1 - 6** you should complete your results table to show the volume of **FA 4** used, the volume of distilled water used and the time taken to the **nearest second** for the blue-black colour to appear.

Experiment 2

- Fill the other burette with distilled water.
- Run 10.00 cm^3 of **FA 4** into a 100 cm^3 beaker.
- Run 10.00 cm^3 of distilled water into the same beaker.
- Use the measuring cylinder to place the following in a second 100 cm^3 beaker.
 - 10 cm^3 of **FA 5**
 - 20 cm^3 of **FA 6**
 - 10 cm^3 of starch indicator
- Add the contents of the second beaker to the first beaker and start timing.
- Stir the mixture once and place the beaker on the white tile.
- Stop timing when a blue-black colour appears.
- Wash both beakers.

Experiments 3 - 6

Carry out **four** further experiments to investigate the effect of changing the concentration of $\text{Fe}^{3+}(\text{aq})$ by altering the volume of aqueous FeCl_3 , **FA 4**, used.

You should not use a volume of **FA 4** that is less than 6.00 cm^3 and the total volume of the reaction mixture must always be 60 cm^3 .

Experiment	Vol. of FA4/ cm ³	Vol. of water/ cm ³	Time taken/s
1	20.00	0.00	20
2	10.00	10.00	27
3	7.00	13.00	32
4	13.00	7.00	24
5	15.00	5.00	22
6	18.00	2.00	21

Recording:

- Constructs a table for 6 results. **M14**
- Correct headings and units for FA4, distilled water and time. Volumes / V/vol in cm³ or / cm³ or (cm³), time / t in seconds or / s or (s). **M15**

Precision:

- All times recorded to the nearest second and volumes of FA4 and distilled water to 0.05 cm³. **M16**

Decision:

- Four further experiments chosen with intervals not less than 2 cm³ and no volume less than 6 cm³. At least one volume must be **less than 10 cm³** and **at least one must be more than 10 cm³**. **M17**
- Water added to make total volume of FA 4 and water 20 cm³ in each experiment and no other changes in volume. **M18**

M17 and M18 penalised if students change volume of other reagents other than FA4 and water.

Accuracy:

- Times increase with decrease in volume FA 4.
- Examiner rounds times to nearest second and calculates (time for expt 2) / (time for expt 1) to 2 decimal place. Ratio is compared with that of Supervisor.
Award marks as follows:
if ratio within 0.2 of Supervisor. **M19**

[6]

Calculations

The rate of reaction can be found by calculating the change in concentration of Fe³⁺(aq) that occurred when enough iodine was produced to change the colour of the indicator to blue-black.

Use your data and the equations on page 5 to carry out the following calculations.

Show your working and appropriate significant figures in the final answer to **each** step of your calculations.

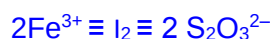
- (b) (i) Calculate the amount of thiosulfate ions, S₂O₃²⁻ used in each experiment.

$$\text{Amount S}_2\text{O}_3^{2-} = \frac{20}{1000} \times 0.0060$$

$$= 1.20 \times 10^{-4} \text{ mol M20}$$

Amount of $\text{S}_2\text{O}_3^{2-}$ [1]

- (ii) Calculate the amount of iron(III) ions, Fe^{3+} , that were used to produce the amount of iodine that react with the amount of $\text{S}_2\text{O}_3^{2-}$ in (i).



$$\text{Amount of Fe}^{3+} = 1.20 \times 10^{-4} \text{ mol M21}$$

OR

$$\begin{aligned} \text{Amount of iodine} &= \frac{1}{2} \times 1.2 \times 10^{-4} \\ &= 0.600 \times 10^{-4} \text{ mol} \end{aligned}$$

$$\begin{aligned} \text{Amount of Fe}^{3+} &= 2 \times 0.600 \times 10^{-4} \\ &= 1.20 \times 10^{-4} \text{ mol M21} \end{aligned}$$

Amount of Fe^{3+} [1]

- (iii) Using your answer to (ii), calculate the change in Fe^{3+} up to the time of appearance of the blue black colour.

$$\text{Change in concentration of Fe}^{3+} = \frac{1.20 \times 10^{-4}}{\frac{60}{1000}} = 0.00200 \text{ mol dm}^{-3} \text{ M22}$$

Change in concentration of Fe^{3+} [1]

- (iv) The following formula can be used as a measure of the 'rate of reaction'.

$$\text{'rate of reaction'} = \frac{\text{change in concentration of Fe}^{3+}(\text{aq})}{\text{reaction time}} \times 10^6$$

Complete the table to show the volume of **FA 4**, the reaction time and the rate in **Experiments 1- 6**. You should include units.

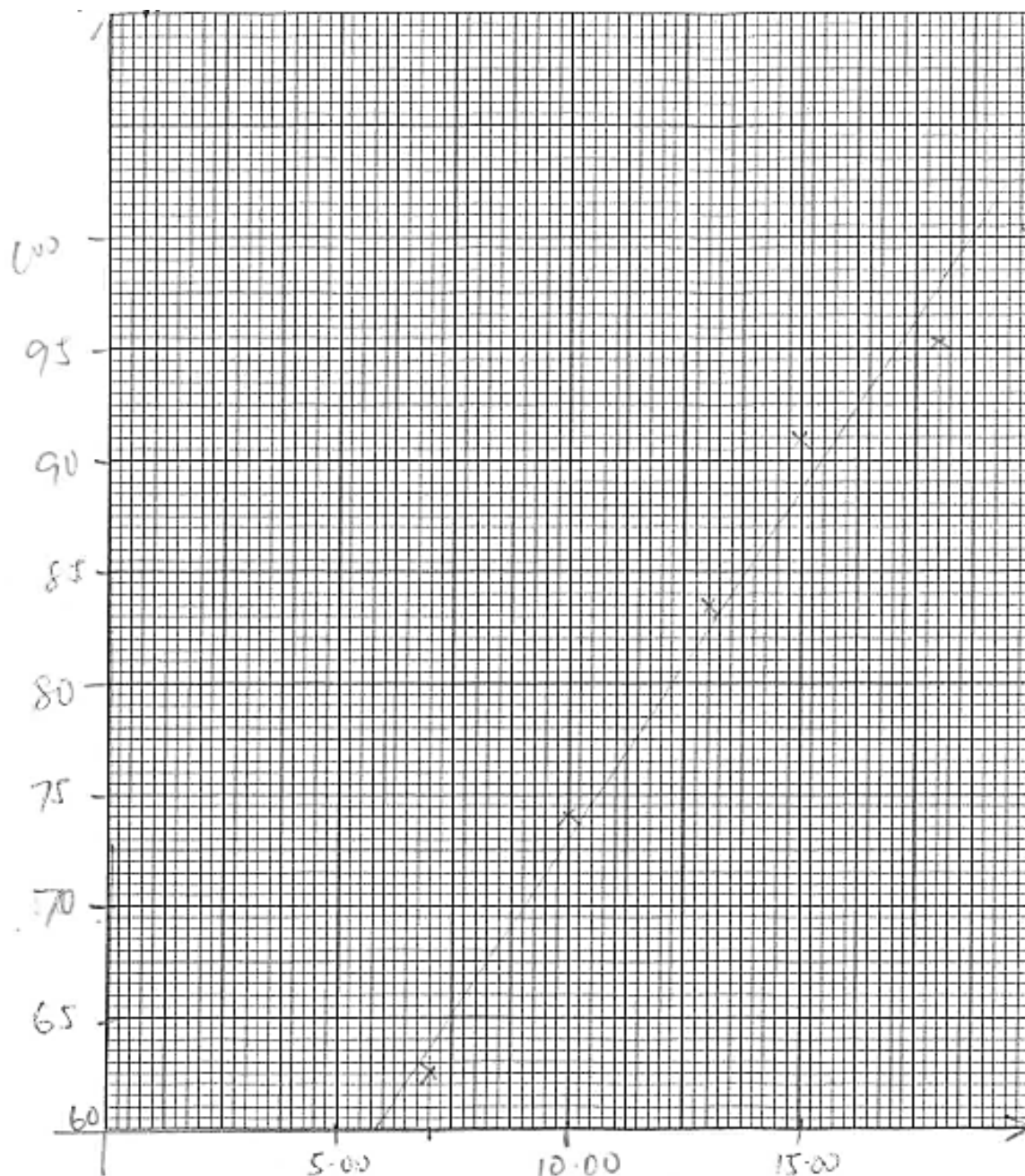
If you **were unable** to calculate a value for the change in concentration of $\text{Fe}^{3+}(\text{aq})$ in (iv), you should assume it is $2.50 \times 10^{-3} \text{ mol dm}^{-3}$. (Note: this is not the correct value.)

Experiment	Vol. of FA4/ cm^3	Time taken/s	Rate/ $\text{mol dm}^{-3} \text{s}^{-1}$
1	20.00	20	100
2	10.00	27	74.0
3	7.00	32	62.5
4	13.00	24	83.3
5	15.00	22	90.9
6	18.00	21	95.2

Rates correctly calculated using $\frac{0.00200}{\text{reaction time}} \times 10^6$

Units for volume given as cm^3 . Units for time given as s. Units for rate given as $\text{mol dm}^{-3} \text{s}^{-1}$, 3 correct columns used and 3 sig fig. Ignore precision for volume of FA4 and time. **M23**
[1]

- (c) On the grid, plot the rate (y-axis) against the volume of **FA 4** (x-axis). Draw a line of best fit through the points. You should identify any points you consider anomalous.



Axes labelled – rate/ $\text{mol dm}^{-3} \text{s}^{-1}$ on y-axis and volume or FA 4 / cm^3 on x-axis. Scales to use at least half of each axis including 0.0 if point plotted. **M24**

Correct plotting – **all** points recorded plotted and within half a small square and within correct small square. Scale must be drawn correctly. **M25**

Draws a line of best fit (can be straight line or curve). Straight lines must be straight (single line with no kinks, drawn using a ruler) or a smooth curve (gradual change in gradient). Points not on the line must be balanced on either side of the best fit line but any points ringed or labelled as anomalous should be ignored. **M26**

[3]

- (d) Using your graph, what conclusion can you reach about the effect of changing the concentration of FeCl_3 on the rate of the reaction between $\text{Fe}^{3+}(\text{aq})$ and $\text{I}^{-}(\text{aq})$?

Rate increases as concentration of Fe^{3+} increases/ rate directly proportional to concentration **M27** since either straight line that pass through the origin shows rate is directly proportional to concentration or gradient is positive **M28** shows that rate increase with increase in concentration (or other words to that effect).

[2]

- (e) It was found, by carrying out experiments similar to those used in (a), that increasing the concentration of I^{-} increased the rate of the reaction.

The student suggested a modification to the method by using the same volumes of all reagents but with the concentration of **FA4** and **FA 5** being doubled their original values. State what the effect would be on the **reaction time** in Experiment 1 and explain how this change would affect any **possible errors** in the measurements.

Reaction time less/shorter. (Less accurate) since larger % error (in time). **M29**
Not "Reaction time faster".

[1]

- (f) Which of the experiments you carried out in (a) had the greatest percentage error in the reaction time? Calculate this percentage error. Assume that the error in measuring the reaction time is ± 0.5 s.

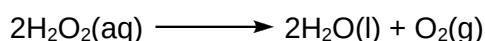
Experiment 1.

$$\text{percentage error} = \frac{0.5}{20} \times 100\% \quad \text{M30}$$

[1]

- (g) **Planning**

Aqueous hydrogen peroxide decomposes into oxygen gas and water. The reaction is normally very slow but is catalysed by solid manganese (IV) oxide.



You are to plan an experiment to investigate how the rate of the catalysed decomposition of aqueous hydrogen peroxide depends on its concentration.

- (i) The rate of decomposition depends on the number of hydrogen peroxide molecules present in a given volume of solution.

Use this information to predict how the rate of decomposition of the hydrogen peroxide depends on the concentration.

Prediction

Prediction: Predicts any direct proportionality e.g. Increasing concentration increases rate or doubling the molecules per unit volume (concentration) doubles the rate. **M31**

- (ii) You are to design a laboratory experiment to test your prediction in (g) (i).

In addition to the standard apparatus present in a laboratory you are provided with the following materials.

- 0.1 mol dm⁻³ aqueous hydrogen peroxide
- A supply of manganese (IV) oxide

Complete the table below to show how you would prepare five solutions of aqueous hydrogen peroxide. Make sure that the correct units are recorded.

expt No.	volume of H ₂ O ₂ / cm ³	volume of H ₂ O / cm ³	concentration of H ₂ O ₂ / mol dm ⁻³
1	40.00	0.00	0.100
2	35.00	5.00	0.0875
3	30.00	10.00	0.0750
4	25.00	15.00	0.0625
5	20.00	20.00	0.0500

Whole table completed. Correct units required for each column. 0 cm³ of H₂O₂ not accepted. Total volume of solutions need not be constant. **M32**

Pre-calculation

Assuming burette is used, maximum volume of O₂ that can be collected will be 50 cm³.

Amount of O₂ = 50/24000 = 2.0833 x 10⁻³ mol

Amount of H₂O₂ = 2 x 2.0833 x 10⁻³ = 4.1667 x 10⁻³ mol

Volume of original H₂O₂ = (4.1667 x 10⁻³) / 0.1
= 41.67 cm³

Note: If gas syringe is used in planning, maximum volume of original H₂O₂ will be around 80 cm³.

- (ii) Give a step by step description of how you would carry out **one** complete experiment.

1. Using a burette, introduce 35.00cm³ of H₂O₂ solution into a 100 cm³ conical flask.
2. Using another burette, introduce 5.00 cm³ of distilled water into the same conical flask.
3. Attach a small tube containing some solid manganese (IV) oxide with a string and stopper it.
4. Record initial volume reading of the syringe.
5. Let the manganese (IV) oxide comes in contact with the solution by loosening the stopper.
6. Start the stopwatch immediately. Swirl to mix gently.
7. Stop the stopwatch once 20 cm³ of gas has been collected.

Give marks for shaking to react the catalyst and solution present. If a thistle or dropping funnel used the mark is for adding the liquid reagent to the catalyst. In other situations give marks for adding solid catalyst to the solution and closing the vessel. **M33**

Start the reaction and a stopwatch simultaneously. **M34**

Recording the time taken to produce a chosen/fixed volume of gas. **M35**

[3]

- (iii) State a problem which might be experienced by someone having to carry out these experiments alone.

Starting the reaction and the stopwatch simultaneously is difficult. Accept any reaction starting process in conjunction with starting the clock. Accept closing the apparatus and starting the clock. **M36**

[1]

[Total: 23]

3 Qualitative Analysis

At each stage of any test you are to record details of the following:

- colour changes seen
- the formation of any precipitate
- the solubility of such precipitates in an excess of the reagent added

Where gases are released they should be identified by a test, **described in the appropriate place in your observations.**

You should indicate clearly at which stage of the test a change occurs.

Marks are **not** given for chemical equations.

No additional tests for ions present should be attempted.

If any solution is warmed, a boiling tube MUST be used.

Rinse and reuse test-tubes and boiling tubes where possible.

Where reagents are selected for use in a test, the name or correct formula of the element or compound must be given.

- (a) **FA 7** and **FA 8** are mixtures. Each mixture contains one cation and two anions from those listed on pages 18 and 19.

Carry out the following tests and record all your observations in the table below.

FA 7 is $\text{Al}_2(\text{SO}_4)_3 + \text{NaCl}$; **FA 8** is $\text{MgCO}_3 + \text{KI}$; **FA 9** is $(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2$

Test	Observations	
	FA 7	FA 8
(i) Place two spatula measure of FA 7 into a boiling tube, then add approximately 5 cm³ of dilute nitric acid using the measuring cylinder. You are to repeat the steps for FA 8. Then	FA 7 dissolves/ A colourless solution is formed. [✓]	FA 8 dissolves and effervescence seen. Gas produced gives a white ppt. with $\text{Ca}(\text{OH})_2(\text{aq})$. [✓]
add about 5 cm³ of distilled water and shake to mix the solution. Use a 1 cm depth of the solution obtained in a test-tube for each of tests (ii) – (v).	No observation required.	
(ii) Add aqueous sodium hydroxide. Then add dilute nitric acid dropwise until in excess.	White ppt. formed, soluble in excess [✓] White ppt. re-forms, soluble in excess $\text{HNO}_3(\text{aq})$ [✓]	Do not carry out this test.
(iii) Add aqueous ammonia.	White ppt. formed, insoluble in excess [✓]	Do not carry out this test.
Test	Observations	
	FA7	FA8
(iv) Add aqueous barium nitrate followed by dilute nitric acid.	White ppt. formed, insoluble in dilute HNO_3 [✓]	No ppt or no observable change [✓]

(v) Add aqueous silver nitrate followed by aqueous ammonia.	White ppt formed, soluble in $\text{NH}_3(\text{aq})$ [✓]	Yellow ppt formed, insoluble in $\text{NH}_3(\text{aq})$ [✓]

[4]

2 [✓] – [1], max [4] M37, M38, M39, M40

(vi) Use your observations to identify the following ions.

FA 7 contains cation

FA 7 contains anions and

FA 8 contains anions and

State and explain all the evidence for your identification of the cation in FA 7.

M41 cation is Al^{3+} and while ppt with $\text{NH}_3(\text{aq})$ insoluble in excess, white ppt with NaOH soluble in excess.M42 FA 7 anions: SO_4^{2-} and Cl^- M43 FA 8 anions: CO_3^{2-} and I^-

[3]

(b) FA 9 contains two cations from the list on page 18.

(i) Transfer approximately half of the FA 9 into a hard-glass test-tube and heat gently at first, and then strongly, until no further change is seen.
Test with litmus papers while you are heating.
Record all your observations below.

Any two observations correct M44

- FA 9 is (pale) green
- steam/ vapour/ condensation/ water liquid
- damp litmus paper turns blue
- yellow/ white/ brown residue formed
- white smoke (produced on strong heating)
- damp litmus turns red on strong heating

[1]

(ii) Carry out further tests that will enable you to identify **both** cations in FA 9.
Describe your tests briefly and state your observations.

With NaOH or NH_3 , dirty green ppt formed which turns brown on contact with air and Fe^{2+} identified. M45With $\text{NaOH}(\text{aq})$ and heat, pungent gas (NH_3) turns damp red litmus paper blue and NH_4^+ identified. M46FA 9 contains cations Fe^{2+} and NH_4^+

[2]

(c) Planning

A solder is an alloy of metals which is used to join other metal pieces together.

A specialist solder that can be used to join together pieces of aluminium is made from a mixture by mass of 65% zinc, 20% aluminium and 15% copper.

You are to plan an experimental procedure to confirm the composition of a powdered sample of this solder, by adding reagents and then extracting from the mixture each of the following in sequence;

- (i) the copper metal,
- (ii) the aluminium as aluminium hydroxide,
- (iii) the zinc as zinc hydroxide

You are provided with

- a sample of this solder, with approximate mass 4 g,
- 1.00 mol dm⁻³ sulfuric acid,
- 1.00 mol dm⁻³ ammonia

No other reagents should be used. Standard laboratory equipment is available including a balance, accurate to two decimal places.

- (i) Complete the flowchart below to show the order in which the reagents would be added to the solder to allow you to extract and separate the components as copper metal, **(Step 1)**, aluminium hydroxide, **(Step 2)**, and zinc hydroxide, **(Step 3)**. You are reminded that aqueous ammonia contains both the base OH⁻ and the complex-forming molecule NH₃.

Step 1	Step 2	Step 3
reagent(s) added	reagent(s) added	reagent(s) added
sulfuric acid [✓]	(excess, aqueous) ammonia [✓]	sulfuric acid [✓]
substance(s) present at the end of the reaction	substance(s) present at the end of the reaction	substance(s) present at the end of the reaction
zinc sulfate [✓] and aluminium sulfate [✓] (copper)	zinc (tetra) ammine [✓] (ions) (aluminium hydroxide)	ammonium sulfate [✓] (zinc hydroxide)
substance(s) removed by filtration (if any)	substance(s) removed by filtration (if any)	substance(s) removed by filtration (if any)
copper [✓]	aluminium hydroxide [✓]	zinc hydroxide [✓]

Allow: Correct formulae or ions instead of names.
2[✓] – 1 mark; max [5] M47, M48, M49, M50, M51

Substances present for each step is conditional based on correct reagent added.

[5]

- (ii) For some of the steps in the procedure you would need to be careful to add an appropriate quantity of a reagent.
For each step of your procedure explain why particular quantities of reagent should be chosen.

Step 1

Sufficient/ enough/ excess sulfuric acid to dissolve the zinc and aluminium **M52**

Step 2

Sufficient/ enough/ excess aqueous ammonia (to precipitate aluminium hydroxide) and to completely dissolve the zinc hydroxide **M53**

Step 3

Sufficient/ enough sulfuric acid to:

Neutralise ammonia **M54 OR**

(re)precipitate the zinc hydroxide but not so much that the zinc hydroxide reacts/ dissolves **M54**

[3]

- (iii) If the mass of aluminium hydroxide obtained was 1.50 g, calculate the mass of aluminium that was present in the solder.

[Ar: Al, 27.0; O, 16.0]

$$\text{Mass of Al in solder} = \frac{27}{27 + 17 \times 3} \times 1.50 \text{ g} = \underline{0.519 \text{ g}} \text{ M55}$$

[1]

[Total: 19]